

Fig. 2.16 Logic Symbol of XOR Gate

2.9.9 XNOR Gate

The XNOR (exclusive - NOR) gate is a combination XOR gate followed by an inverter. Its output is "true" if the inputs are the same, and "false" if the inputs are different. In simple words, the output is 1 if the input are the same, otherwise the output is 0. The logic circuit of XNOR gate is



Fig. 2.17 Logic Circuit of XNOR Gate

The output of the XNOR is NOT of XOR

$$C = \overline{A \oplus B}$$

$$= \overline{A} \cdot B + A \cdot \overline{B}$$

$$= AB + \overline{A}\ \overline{B}$$

(Using De Morgan's Theorem)

In boolean algebra, $\odot$ or "included dot" stands for the XNOR.

Therefore, $C = A \odot B$

The logical symbol is

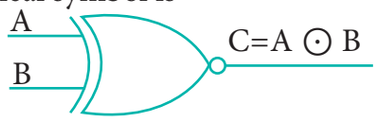


Fig. 2.18 Logic Symbol of XNOR Gate

The truth table for XNOR Gate is

Input		Output
A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

Table 2.14 Truth Table for XNOR Gate

Using combination of logic gates, complex operations can be performed. In theory, there is no limit to the number of gates that can be arranged together in a single device. But in practice, there is a limit to the number of gates that can be packed into a given physical space. Arrays of logic gates are found in digital integrated circuits.

Theorems of Boolean Algebra

Identity
- $A + 0 = A$
- $A \cdot 1 = A$

Complement
- $A + \overline{A} = 1$
- $A \cdot \overline{A} = 0$

Commutative
- $A + B = B + A$
- $A \cdot B = B \cdot A$

Associative
- $A + (B + C) = (A + B) + C$
- $A \cdot (B \cdot C) = (A \cdot B) \cdot C$

Distributive
- $A \cdot (B + C) = A \cdot B + A \cdot C$
- $A + (B \cdot C) = (A + B) \cdot (A + C)$

Null Element
- $A + 1 = 1$
- $A \cdot 0 = 0$

Involution
- $\overline{(\overline{A})} = A$

Idempotence
- $A + A = A$
- $A \cdot A = A$

Absorption
- $A + (A \cdot B) = A$
- $A \cdot (A + B) = A$

3rd Distributive
- $A + \overline{A} \cdot B = A + B$

De Morgan's
- $\overline{A + B} = \overline{A} \cdot \overline{B}$
- $\overline{(A \cdot B)} = \overline{A} + \overline{B}$

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